[INCH-POUND]
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SUPERSEDING
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21 January 1975
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15 August 1964

COMMERCIAL ITEM DESCRIPTION

HEATERS, VEHICULAR COMPARTMENT 28 VOLT, DC

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE. This CID covers two types of heaters of 28 volt direct current (DC) nominal rating, used for heating crew and engine compartments of military vehicles or equipment. Type I heaters are intended for use on military vehicles or equipment which may be required to operate in an ambient temperature to minus 65 degrees Fahrenheit (-65°F); from which a supply of gasoline is available; when heated air is required without the vehicle engine being operated. Type II heaters are intended for use on military vehicles or equipment from which a supply of heated circulating liquid (engine coolant) is available, which may be required to operate in an ambient temperature to -25°F.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent by letter to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

FSC 2540

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1.1 Classification. Heaters shall be of the following types and capacities, as specified (see 6.2):

Type I - Gasoline Fired:
Capacity A - 20,000 British thermal units per hour (Btu/hr).
Capacity B - 30,000 Btu/hr.
Capacity C - 60,000 Btu/hr.

Type II - Liquid Coolant Operated:
Capacity B - 30,000 Btu/hr.

2. SALIENT CHARACTERISTICS

2.1 Materials. Materials for type I heaters shall be as specified in the applicable engineering drawing (AED). Materials for type II heaters shall be in accordance with the manufacturer's material specifications for heaters. The use of recovered material made in compliance with regulatory requirements is acceptable providing all requirements of this CID are met (see 4.1).

2.2 Design and construction. Unless otherwise specified, type I heaters shall fall within the envelope outline of the AED and shall conform to the detail requirements for mounting dimensions, external electrical connections and fuel inlet. Type II heaters shall be constructed in accordance with figures 1, 2, and 3. Type I and type II heaters shall meet the applicable requirements of Section 393.77, Motor Carrier Safety Regulations (MCSR) in effect on date of manufacture.

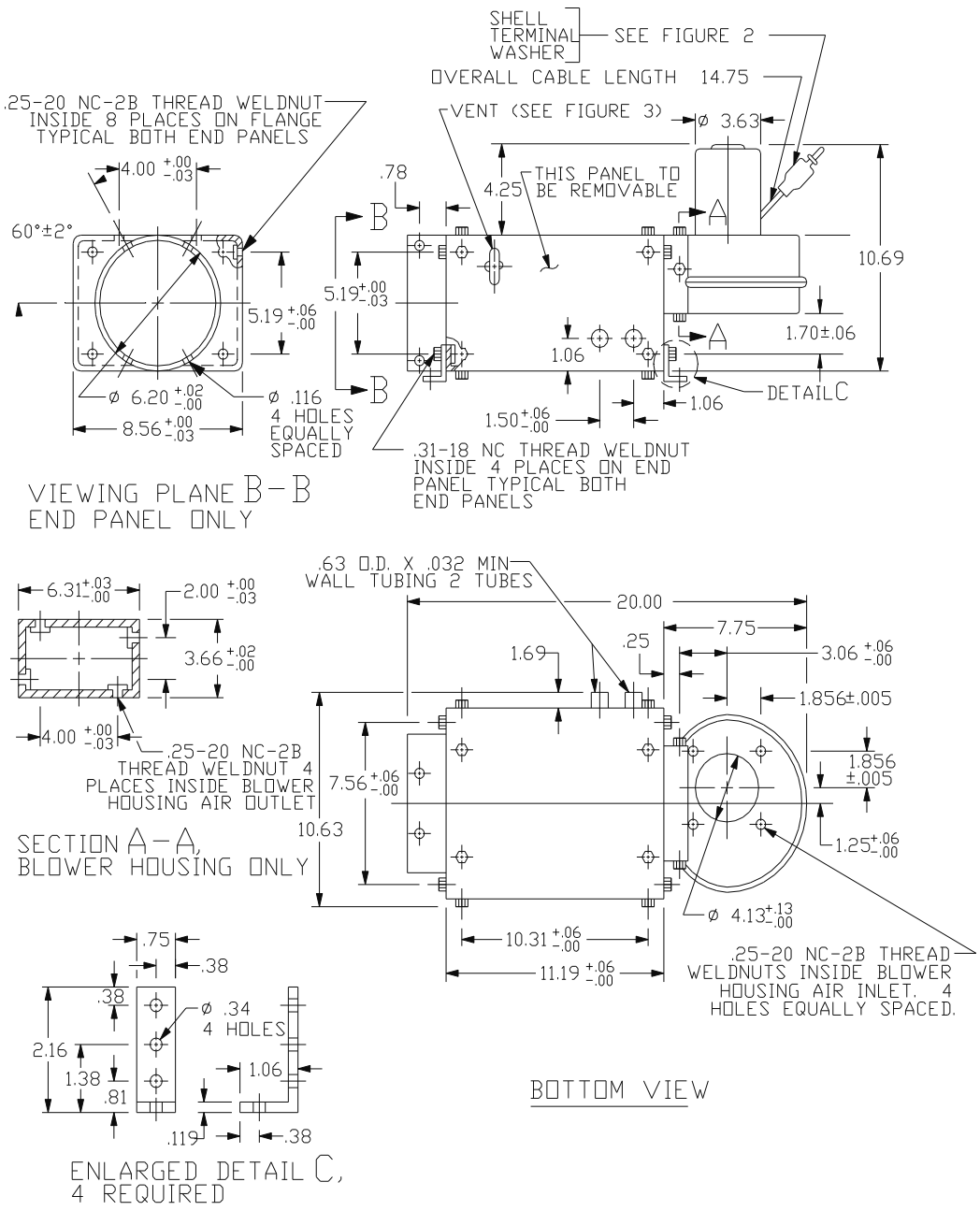
2.2.1 Interchangeability. All parts having the same manufacturer's or Army part numbers shall be completely interchangeable with each other with respect to installation and performance.

2.3 Controls. When specified (see 6.2), heaters shall be furnished complete with controls to start, stop, and operate heaters.

2.4 Coolants. Type II heaters shall perform as specified herein using the following type coolants/rust inhibitors:

a. Water and antifreeze compound conforming to A-A-870.
b. Arctic type antifreeze compound of a single phase homogeneous mixture. This compound shall provide protection against freezing in regions where the ambient temperature remains close to -40°F for extended periods of time, but may drop to as low as -90°F.
c. Water and corrosion inhibiting compound consisting of the chemical composition shown in table I.

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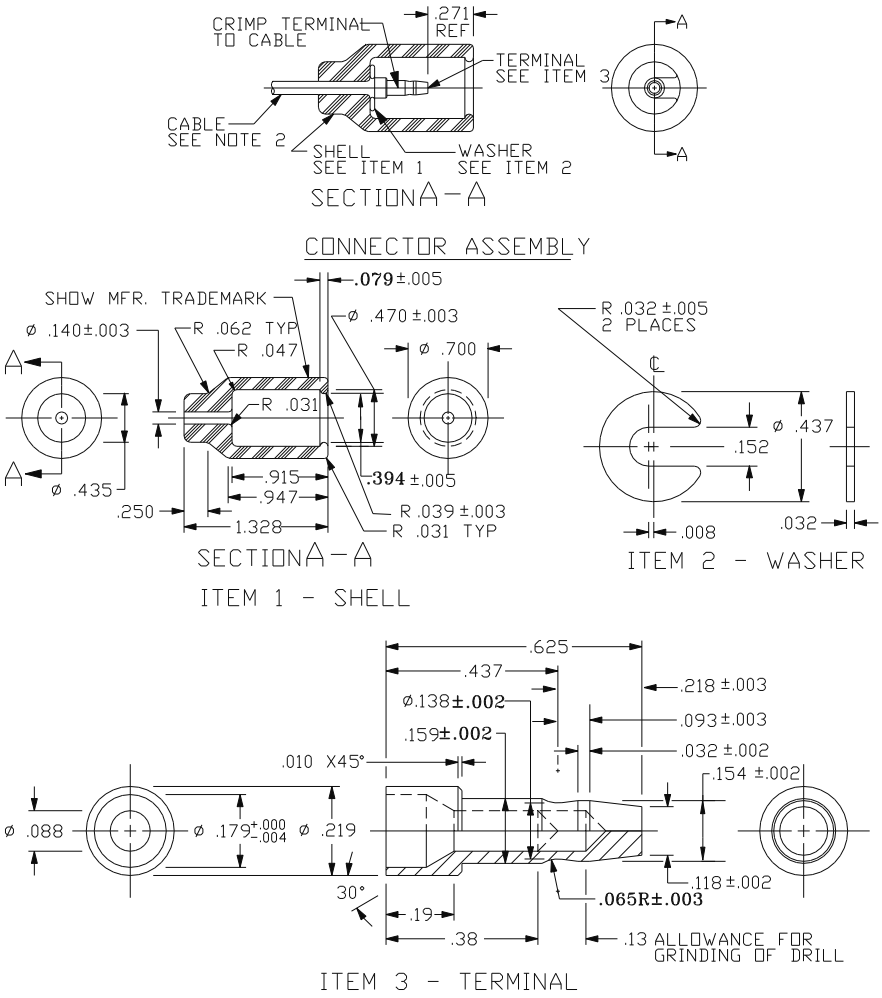


NOTES:

1. Dimensions are in inches, maximum, except detail C, unless otherwise specified. Detail C tolerances are ± .06.
2. Basic assembly is shown. Heater may be adapted to various configurations by rearranging or interchanging end panels or blower housing attachment.

FIGURE 1. Type II heater.

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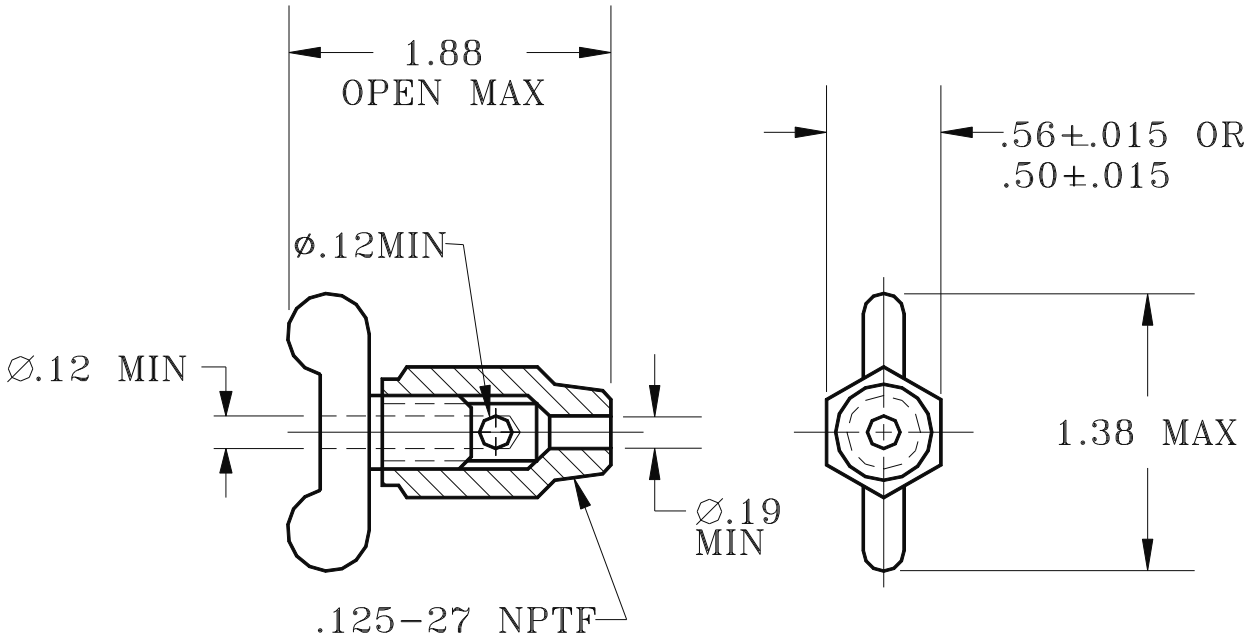


NOTES:

1. Dimensions are in inches. Unless otherwise specified tolerances are ± .010 for shell, ± .003 for washer and ± .007 for terminal.
2. The cable shall be unshielded single conductor type, 14 AWG size, with outside diameter of .160 ± .010. The cable shall be designed and constructed as shown in A-A-52499.
3. Shell material shall be rubber grade M5BC610A14C12E034F19 or equivalent as specified in ASTM D2000.
4. Shell shall be marked “OZ” (ozone resistant).
5. Washer material shall be brass, temper ½ hard minimum.
6. Finished washer shall have burrs and sharp edges removed.
7. Terminal material shall be brass, temper ½ hard minimum.
8. Terminal finish shall be silver plate, .0002 minimum thickness.

FIGURE 2. Type II heater connector assembly.

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NOTES:

1. Material shall be brass. Handle may be zinc plated steel.
2. Dimensions are in inches.

FIGURE 3. Type II - Heater vent.

TABLE I. Chemical composition.

Components	Weight percent
Distilled or deionized water (total added)	63.4 ± 0.6
Sodium metaborate ($Na_2B_ZO_4$ $8H_2O$)	29.0 ± 0.3
Potassium Silicate solution	4.6 ± 0.2
Sodium mercaptobenzothiazole, 50% agueous solution, by weight	3.0 ± 0.1

2.5 Fuel. Type I heaters shall perform as specified in 2.6.1 using military combat grade automotive gasoline or commercial grade automotive gasoline.

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2.6 Performance.

2.6.1 Type I heater.

2.6.1.1 Heat output. With inlet air temperature at -65 ± 5°F, the type I heaters shall have a heat output as specified in table II.

2.6.1.2 Air discharge rate. The type I heaters when performing at high setting, shall discharge air at not less than the rates specified in table II.

2.6.1.3 Fuel flow. Consumption of gasoline conforming to 2.5 in type I heaters shall not exceed the flow rate values specified in table II.

TABLE II. Type I Heaters.

		Capacity		
		A	B	C
Heat Output: Btu/hr.	High Setting	18,000	27,000	54,000
	Low Setting	7,200	16,200	27,000
Air Discharge: CFM 1/	Unrestricted	80	105	200
	Air Back Pressure	50 at .5 inch H_20	85 at 1.0 inch H_20	150 at 1.0 inch H_20
Fuel Flow: Gal/Hr.	High heat	.29	.48	.83
	Low Heat	.13	.30	.44
Current: Amperes	Start	12	18	21
	Run	4	7	11

1/ = cubic feet per minute.

2.6.1.4 Current. Using a starting voltage of 24.0 ± 0.5 volt DC and a test ambient temperature of -65°F, the current shall be not more than the values specified in table II.

2.6.1.5 Overheat switch. The overheat switch shall break the circuit and cut off the flow of fuel when the ventilating air mixed with ambient air is heated to a temperature of 375 ± 25°F.

2.6.1.6 Cycling. Type I heaters shall start in less than four minutes and purge in less than three minutes when cycled at high setting for 50 cycles at an ambient temperature of -65 ± 5°F.

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2.6.1.7 Endurance. The heater shall operate without a breakdown of the heat exchanger, electrical motor, combustion air blower wheel, ventilating air blower wheel, or fuel control valve during not less than 500 hours of continuous use, operating at ambient temperatures as low as -65°F, alternately at high and low heat settings, for 24 hour periods and cycling 20 minutes "ON" and 10 minutes "OFF".

2.6.1.8 Safety. All heaters shall comply to DOT standards cited in 49 CFR, section 177.834 in effect on date of manufacture.

2.6.2 Type II heaters.

2.6.2.1 Heat output and air discharge. The heat output and air discharge rate shall be not less than values in table III, as determined per SAE J638 procedure.

TABLE III. Type II heat output and air discharge.

Inches H_2O Back Pressure	Heat Output Btu/hr	Air Discharge: CFM
0	28,000	150
1.6	15,000	75

2.6.2.2 Cycling. The heater shall have no coolant leakage from the coolant circulation system when cycled at 20 cycles per minute, pressure rise and fall from 0 to 32 pounds per square inch (psi), for 2500 total cycles. Cycling shall be accomplished at ambient temperature of -25 ± 5°F. Each cycle shall consist of starting at high heat, running for 15 minutes, purging, and cooling the heater for 35 minutes with auxiliary blower.

2.6.2.3 Endurance. The heater shall operate without a breakdown of the heat exchanger, electric motor or fan during not less than 250 hours of continuous operation with a coolant temperature of 180 ± 5°F with zero back pressure, and not less than 250 hours with 1.6 inches of H_2O back pressure. Starting time to full load shall be 10 seconds maximum.

2.6.3 Type I and II heaters.

2.6.3.1 Voltage. The heater shall start and operate satisfactorily at an applied voltage of 20 to 28.5 volts DC.

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2.7 Environmental.

2.7.1 Low temperature operation, type I heaters. Type I heaters shall have operational stability at temperatures to -65 ± 5°F and type II heaters shall have operational stability at temperatures to -25°F.

2.7.2 Vibration. The heaters shall start and operate while vibrating at an amplitude of .25 inch for one hour in the direction of each of the three major axes of the heater. While accelerating and decelerating at a uniform rate for an interval of one minute, the frequency shall vary through a range of 10 to 55 cycles per second. There shall be no evidence of leakage of fuel from the type I heaters, or of coolant from the type II heaters during or after the vibration.

2.7.3 Corrosion resistance. With all duct openings sealed, the heaters shall show no evidence of corrosion that would affect their functioning after being subjected to a 96-hour salt solution spray.

2.7.4 Electromagnetic interference. Unless otherwise specified (see 6.2) or in applicable drawings, the heaters shall meet the electromagnetic interference requirements of products offered commercially.

2.8 Identification and markings. Identification and markings shall be permanent and legible and shall include as a minimum, the manufacturer's identification code (CAGE), the engineering drawing part number (type I heaters) and PIN AA52545-1 (type II heater).

2.9 Nameplate. A nameplate shall be attached to the heater in a location, such that the plate may be easily read when the heater is installed in the vehicle. The location of the nameplate shall be as specified in the contract or ordering data (see 6.2). The nameplate shall contain the following minimum information:

HEATER 28 VOLTS DC
ENGINEERING DRAWING PART NUMBER (TYPE I HEATERS)
PIN AA52545-1 (TYPE II HEATERS)
MANUFACTURER'S IDENTIFICATION
MANUFACTURER'S SERIAL NUMBER

2.10 Painting. The exterior surfaces of the type I heater case and outlet adapter shall be cleaned, degreased, and then painted with paint heat tested to 400°F. Other surfaces of the type I heater, and the exterior surfaces of the type II heater shall be finished as specified in the contract (see 6.2).

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3. QUALITY ASSURANCE PROVISIONS

3.1 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Addresses for obtaining copies of referenced documents.

6.1.1 Government documents. Copies of MCSR Section 393.77 and 49 CFR parts 100 to 177 can be obtained from Superintendent of Documents, Government Printing Office, Washington, DC 20402-9371. Copies of A-A-870 “Antifreeze/Coolant, Engine: Ethylene Glycol, Inhibited Concentrated” and A-A-52499 “Lead, Electrical (Storage Battery)” are available from the Defense Printing Services Detachment Office, Bldg. 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.

6.1.2 Non-Government documents.

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE J638 - Motor Vehicle Heater Test Procedure, Recommended Practice.

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(Applications for copies should be addressed to the Society of Automotive Engineers (SAE), 400 Commonwealth Drive, Warrendale, PA 15096).

6.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this CID.
b. Issue of Department of Defense Index of Specifications and Standards (DODISS) and the specific issue of individual documents referenced.
c. Type and capacity of heater.
d. Title, number, and revision letter of the AED for type I heaters.
e. PIN for type II heaters.
f. Quantity of heaters required.
g. Electromagnetic interference requirements if other than commercially available.
h. Location of nameplate.
i. If controls are to be furnished.
j. Surface finish of heaters.
k. Selection of applicable level of packaging requirements.

6.3 Cross-reference. Heaters conforming to this CID are interchangeable/substitutable with heaters conforming to MIL-H-3199E and MS51326A(AT).

MILITARY INTERESTS:

Custodians:
Army - AT
Air Force - 99

Review Activities:
Navy - MC
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

PREPARING ACTIVITY:
Army - AT

(Project 2540-0428)

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